Daniel **Carrión**

Engineer & Designer

About me

Graduated in electronic engineering, robotics and mechatronics, with a master's degree in mechatronics. I love technology, specifically electronics and SoC programming. In my last years I have acquired programming skills in C/C++ and FreeRTOS with mCU thanks to projects like HortSost in which I got involved as a technician. As a researcher, I delved into the programming of RISC-V architectures and design of applications and wireless sensor network infrastructures framed in IoT projects.

Personal

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Areas of specialization

C/C++ Programming • IoT • PCB Design • 3D Design • Electronic • Robotic & Control

Skills

Coding and Programming
Data Analytics
Data interpretation and Charting
Active listening and Empathy
Adaptability
Continuous learning

Interests

SoC Programming / Automating processes Web Programming / Reverse engineering / Discuss

Languages

Spanish C2 mother tongue
English B2 FCE Cambridge

Short Resumé

01/2025–Present

Test Engineer & Certification Specialist

IoT, Cybersecurity & Compliance · DEKRA

- Participation in the evaluation team responsible for Google pKVM for Android SESIP Level 5 certification, including cryptographic analysis, secure virtualization design review, and generation of evaluation evidence, leveraging automation and custom tooling through Python and C/C++ scripts.
- Execution of IoT device security evaluation projects with Bluetooth (Classic, Low Energy - BLE, Mesh), LTE, and wireless technologies, analyzing attack surface, pairing, encryption, and key management, including test automation and analysis with Python and C/C++ scripts.
- Analysis of critical security functions: secure boot, secure firmware upgrade, secure debug, and secure storage, verifying design, implementation, and resilience against software and physical attacks, developing scripts/tools in Python and C/C++ for testing and evidence collection.
- Device evaluation using I2C, JTAG, BLE, Ethernet, and MACsec protocols, testing domain isolation and Layer 2 traffic protection.
- Daily development and execution of test campaigns in Linux environments, virtual machines, and bash/CMD consoles for log analysis and test infrastructure management, integrating automations and scripts.

02/2022–09/2024

Technical staff & Research personnel

Programmer · COMPUTERS ARCHITECTURE DEPT., UMA

- Carried out programming and conditioning activities for the technical part of the HortSost II project at the University of Malaga.
- Co-tutored final degree projects of students in the department.
- Led the technical adaptation and automation of a greenhouse, handling various sensors, electrical conditioning, programming (ESP32), and data management with MQTT and NODE-RED on UNIX servers.
- Final degree project awarded 2nd place in the Bizintek 22/23 Awards.
- Developed a self-configuring sensor infrastructure using C/C++ and FreeRTOS with ESP32 for edge data processing.
- Published and presented "Auto-configurable wireless sensor network for edge computing" at a national computer science congress.
- Signal conditioning of the A121 radar sensor (60GHz PCR) for biometric monitoring in enclosed spaces.

06/2019–01/2020

Technical Staff

Worker · SITER 2000

- Internship focused on consulting and studies of GSM/DCS/UMTS and FTTH mobile network deployments.
- Analyzed engineering documents and As-builts for access point installations in educational centers.

Degrees

2023 **Master in Mechatronics Engineering**
M.A. · University of Malaga

2023 **Project Management of Photovoltaic Solar Installations: Solar Project Management**
M.A. · EUROINNOVA BUSINESS SCHOOL

2021 **Grad. Electronic engineering, robotics and mechatronics**
B.E. · University of Seville-Malaga

Certificates

2024 **CATIA V5 - Part Design (Ass)**
Cert. · Dassault Systèmes

2020 **PHP, JAVASCRIPT and MySQL developer**
Cert. · DEUSTO FORMACIÓN

2019 **RPAS pilot and radio operator**
Cert. · Aerocamaras

Publication & Talks

Jun. 2024 "Auto-configurable wireless sensor network for edge computing", at: *Spanish Congress of Informatics* in A Coruña, Jun. 2024.